Determining Randomness Using LLMs

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Abstract

LLMs have many uses, but can they be used to distinguish random from not random. Through a series of prompts and tests given to three LLMs, these models had to break down the data given to find patterns or the lack thereof. These models were able to differentiate random from non-random bits for the most part, although the specifics of these tests were inconsistent or incorrect.

1 Introduction

Large Language Models, or LLMs for short, have been growing in popularity and in complexity in recent years. With more and more research and development being done, the accuracy of these LLMs has increased. The main focus of this paper is to test multiple LLMs to see whether they can distinguish random from non-random and to test their accuracy at figuring this out.

1.1 What is "Random"?

Random in cryptology means that every subsequence of a larger binary sequence is uniformly distributed for sequence lengths of k. In respect to this paper and its testing, random is a series of boolean bits (1 or 0) that have no distinct pattern, and are generated using the Python 'random' library.

2 Parameters

In the tests conducted, a few pieces of information are needed to generate the data. Some of those include the biases in each file, and the value for n, the number of bits generated in each file.

2.1 Biases

The dataset created for the tests has ten files. In each of these files, except for the control, is biased. The biases for each file are shown in *Table 1.*

2.2 Determining n

To determine which n value will be used, the LLMs were prompted to detect randomness for each dataset. These preliminary results showed that $n = 2^{20}$ would be the best for the tests as it provides a large enough size for patterns to be detected, and its size is within reason.

3 Experimentation

For all of the following data collected, the value of $n = 2^{20}$ was used for the number of bits generated in each file. This was then converted to hexadecimal to conserve space in the files. The ten files generated were then given to three LLM models,

ChatGPT, Claude, Gemini

to analyze for randomness. The first test prompted to the LLMs was the χ^2 test.

File	Bias %	Description
file0.txt	None	Control, no bias.
file1.txt	60%	Every third bit is biased
file2.txt	55%	Every third bit is biased
file3.txt	80%	Every 2, 4, 6, 8, ... bits are biased
file4.txt	80%	Every 3, 5, 7, 9, ... bits are biased
file5.txt	55%	Every 2, 4, 6, 8, ... bits are biased
file6.txt	55%	Every 3, 5, 7, 9, ... bits are biased
file7.txt	55%	Every third byte is manipulated; every bit is biased
file8.txt	55%	Every 2, 4, 6, 8, ... bytes are manipulated; even bits are biased toward 0 and odd bits toward 1
file9.txt	55%	Every 1, 3, 5, 7, ... bytes are manipulated; even bits are biased toward and odd bits toward 1

Table 1: File names and biases to data.

3.1 χ^2 Tests Results

For these tests, the LLMs were prompted to only use χ^2 tests to determine randomness.

3.1.1 [1] ChatGPT

ChatGPT determined that the control file was random, and that files 1-9 all had χ^2 value and p values that would indicate non-randomness.

3.1.2 [2] Claude

Claude also found that the control file was random, and that files 1-9 had χ^2 values and p values that indicate non-randomness.

3.1.3 [3] [4] Gemini

Gemini did not correctly identify which files were random, and which were not. It found that files 0, 1, and 2 were random, and that files 3-9 were not random. The three files that it determined to be random all had χ^2 values well below the maximum, and p-values close to 1. The other files had values well above both the χ^2 and p-value maximums.

3.2 Other Tests

For these test, the LLMs will not be limited to only χ^2 test and be allowed to do which ever test it wants.

3.2.1 [5] ChatGPT

For files 0 - 9, ChatGPT accurately detected irregularities in the data.

- file0.txt - it determined that it was random by calculating Entropy (7.9986 /8), Ones ratio (50.006%), and χ^2 test
- file1.txt - it determined that the data was not random; however, it was not able to find the pattern. It argued that there was a "Strong over-representation of 0, 1, 2, 4, 8 which is reasonable since this test biased the data towards more zeros but was not able to figure out the positions where bias was more likely.
- file$2, 3, 4, ..., 9$.txt - Using the ratios of ones and also calculating the entropy levels, it was able to detect irregularities in files $2, 3, 4, ..., 9$.

3.2.2 [6] Claude

Claude did a good job at detecting irregularities. It had a similar process to ChatGPT.

- file0.txt - Measured Entropy at 7.999/8, calculated mean value of 127.4, and found χ^2 test had value of 248
- file1.txt and file2.txt - Gave similar reasoning for file 1 and file 2. Measured Entropy at 7.706/8 and 7.763/8 respectively, files had mean values of 102 and 104, and found that both files showed unequal distribution in the least significant 4 bits of a byte
- file3.txt and file5.txt - Found that both files showed an odd-bit pattern, with odd bits (1,3,5,7) being ∼90% set in file3.txt and ∼77.5% set in file5.txt, while even bits remained ∼50%. Measured Entropy at 5.885/8 and 7.079/8, with mean values of 195 and 174 respectively. Claimed that file3.txt always had an 0xAA mask and file5.txt being a "weaker" version of the same pattern
- file4.txt and file6.txt - Found that both files were bias toward 1s, with all bits ∼72% set in file4.txt and ∼65% set in file6.txt. Measured Entropy at 6.817/8 and 7.467/8, mean values of 183 and 166 respectively
- file7.txt, file8.txt, and file9.txt - Claimed that all three had nearly identical extreme results. Measured Entropy at ∼1.536/8, with 87.6% of bytes being 0x00 and declared that nonzero bytes were appearing only in the first 16KB

3.2.3 [7] Gemini

Gemini tested the data by evaluating the distribution and frequency of each hexadecimal character. It also would occasionally check the frequency of zeros and look for unusual binary patterns
It was able to detect that file0.txt was random and files 1-9 were not random.

4 Analysis

The χ^2 tests done showed mixed results. While ChatGPT and Claude were able to accurately identify which files were random/not random, Gemini struggled to figure this out. Initially, it did not give any value associated with the χ^2 test, but after prompting it for those values, it gave them. For the values given by ChatGPT and Claude, they were very similar in relative distribution, only with the much higher values being extremely. Gemini's values were comparatively extremely low and did not have similar ratios to either other LLMs.

When it comes to the prompting the LLMs to determine randomness in other ways, the results were pretty similar. All of them found that file 0 was the control file. They all found that files 1-9 were not random, and had some interesting finding. Both ChatGPT and Claude saw that files 1, 2, 7, 8, and 9 had "Low Bit Sparse Structured Data" or low bias. Both also caught that files 3, 4, 5, and 6 had "High Bit Saturated Structured data", or high to mild bias. Gemini found that files 1-6 had high bias and files 7-9 had low bias with many trailing zeros.

5 Conclusion

LLMs provide mixed results when trying to determine randomness. LLMs like ChatGPT and Claude are relatively good at identifying whether a set of data is random or not using a χ^2 test, where Gemini fails at doing this. Although these LLMs can distinguish random from non-random, the results from these χ^2 tests were very different, making their results unreliable. When it comes to identifying randomness and patterns in data, this is where the LLMs are more reliable. All of the LLMs were able to identify which datasets were random and not random, and ChatGPT and Claude were able to pick up on some of the patterns. Overall, LLMs can be used to differentiate random bits from non-random bits; however, any more specific tests or patterns requested would provide mixed or inaccurate results.

6 Appendix

6.1 Preliminary Results

First test, n = 2^5
Claude detects that n is far too small for a proper χ^2 Test

Second test, n = 2^{10}
Still too small to get decent results from LLMs.

Third test, n = 2^{15}
Interesting result. Only one file, every 10th bit was manipulated (51% chance of being 0). Claude thought every 1st, 3rd, 5th, 7th bit was manipulated.

References

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